

Communication-free Cooperative Transportation with Multiple UAVs

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Abstract: This paper investigates the cooperative transportation problem using multiple UAVs equipped with a cable-suspended point-mass load. Instead of using inter-UAV communication, our system relies only on locally available information. We introduce a distributed communication-free control strategy to stabilize an evenly spaced circular formation by leveraging cable tension. A leader-follower approach is designed to ensure load position tracking while maintaining the desired formation. The proposed strategy has been validated through extensive numerical simulations and real-world experiments with 3 UAVs carrying a small load.

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1. INTRODUCTION AND RELATED WORK

With the significant development in the field of Unmanned Aerial Vehicles (UAVs), aerial transportation has attracted attention from researchers due to the variety of applications in construction, package delivery, and more. Designing a system that is adaptable to the diverse shapes and weights of transported loads is a real challenge since UAVs suffers from limited load-lifting capabilities. A common solution to the aforementioned problem is to employ several UAVs to do the transportation task.

Several studies, such as in Lee (2014), and Li et al. (2024), have considered the modeling of a group of UAVs carrying a load, where the system is controlled by a centralized geometric PID controller. To cope with the scalability limitations of centralized approaches, researches have proposed decentralized control strategies for multi-UAVs cooperative transportation. Shirani et al. (2019) developed a decentralized position controller for each UAV to keep a desired formation around the load, and to track a desired trajectory for the load. García de Marina and Smeur (2019) have used inter-UAV distances to maintain the UAVs in a desired formation, rejecting the effect of the tension using dynamic inversion controller, and allowing a desired acceleration tracking of the formation by introducing disagree on the inter-distances. Both studies assume access to the states of neighboring UAVs, which in real tests requires digital communication.

In non-ideal world, digital communication suffers from issues like latency, packet loss, and high power consumption. To reduce the burden of digital communication, Zhang et al. (2021) developed a control scheme with a self-

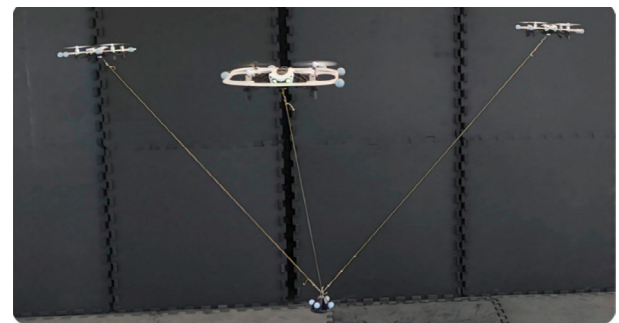


Fig. 1. Three mini-UAVs cooperatively carrying a load using cables (video: <https://youtu.be/ORG0gBFamWA>).

triggering strategy that obtains the states of neighboring agents only when needed.

Other researchers explored the possibility of completely eliminate the need of communication by exploiting the contact force between the robot and the transported object. Collaborative manipulation and transportation *without explicit communication* between the robots can be achieved by knowing or sensing these contact forces. In Wang and Schwager (2016), a decentralized collaborative strategy was introduced using a leader-follower scheme for ground robots transporting a load, with a single leader responsible of guiding the fleet to a desired position and an arbitrary number of followers controlled using a force-feedback controller.

Gabellieri et al. (2020) have developed a general collaborative load-manipulation control strategy using a leader-follower scheme for any type of agents (ground, aerial, and underwater robots); an admittance controller is employed for all agents, where the leader exerts force on the load to achieve a desired position and orientation, and the followers maintain a desired force in the link with the load to retains a desired orientation. To implement this strategy in a real setup, it is essential to have precise estimation of

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the load's weight and orientation, which highlights a significant practical challenge in aerial transportation. To tackle this challenge, the authors have considered the particular case of two UAVs transporting a bar-shaped load and studied several implementation challenges, in Gabellieri et al. (2023) and the references therein. The case of two UAVs transporting a bar load has also been explored by Tagliabue et al. (2017), Tagliabue et al. (2019) and Wu et al. (2023).

In this study, we address the challenge of cooperative aerial transportation using multiple UAVs by considering a point-mass model for the load to eliminate the need for orientation control. Additionally, we decouple the altitude control from the two-dimensional transportation task. Our primary objective is to achieve cooperative aerial transportation without explicit communication and without extensively relying on the global position of each UAV. We introduce a decentralized communication-free tension-based formation controller that is able to stabilize an arbitrary number of UAVs transporting a payload.

This paper is organized as follows: Section 2 formulates the problem at hand and introduces the model used for the study. Section 3 presents the leader-follower control scheme and describes the design of the various controllers implemented. Section 4 and Section 5 validate our control strategy through both realistic simulations and real-world tests. Finally, Section 6 concludes the study.

2. PROBLEM FORMULATION

We consider a team of N aerial robots attached to a load by a rigid cable of length l . The mass of the load m_L is known. Fig. 2 illustrates the considered system and the variables involved. Let $\mathcal{F}_W = \{x_W, y_W, z_W\}$ denote the global inertial frame with z_W facing the opposite direction of the gravity vector, and $\mathcal{F}_{Ri} = \{x_{Ri}, y_{Ri}, z_{Ri}\}$ denote the body frame attached to the Center of Mass (CoM) of each UAV $i = 1, \dots, N$. The configuration of each robot is described by the position vector $\mathbf{p}_i = [x_i, y_i, z_i]^T \in \mathbb{R}^3$ and the orientation of $\mathcal{F}_{Ri}$ w.r.t. $\mathcal{F}_W$ described by the rotation matrix ${}^W_{Ri} \mathbf{R}_i(\phi_i, \theta_i, \psi_i) = \mathbf{R}_i$ obtained from the Euler angles roll, pitch, and yaw respectively.

2.1 UAV dynamics

For rotary wing UAVs that can generate thrust only in z_{Ri} direction, we assume that the internal dynamics are well controlled. Thus, the inputs to control the acceleration of the UAV i are $\mathcal{U}_i = [\phi_i, \theta_i, \psi_i, f_i]^T$, where ϕ_i, θ_i, ψ_i are the Euler angles, and f_i is the thrust magnitude. Note that changing $\mathcal{U}_i$ is equivalent to changing the thrust vector $\mathbf{F}_i \in \mathbb{R}^3$ expressed in $\mathcal{F}_W$, therefore the simplified dynamical model of UAV i in free flying is:

$$\begin{aligned} \dot{\mathbf{p}}_i &= \mathbf{v}_i \\ m_i \ddot{\mathbf{p}}_i &= \mathbf{F}_i - m_i g \mathbf{z}_W \end{aligned} \quad (1)$$

where $\mathbf{v}_i \in \mathbb{R}^3$ denotes the linear velocity of UAV i , m_i its mass, the thrust $\mathbf{F}_i = \mathbf{R}_i f_i \mathbf{z}_{Ri}$, and $\mathbf{z}_W = [0 \ 0 \ 1]^T$.

2.2 Tension model

Estevez et al. (2024) have reviewed the different methods used for modeling the connection between a UAV and a

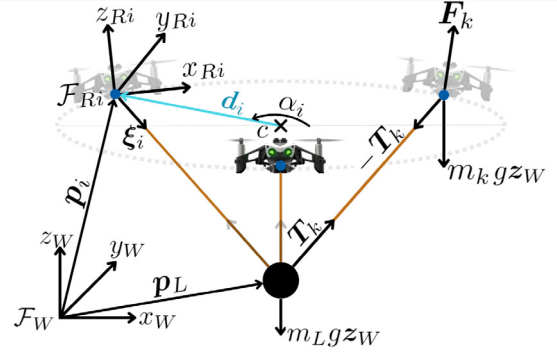


Fig. 2. Representation of the system and its major variables. The system under consideration of the general case of any number of aerial vehicles transporting a point-mass load.

suspended load. It is common in the literature to assume that the gravitational force acting on the load keeps the cable taut. Therefore, in this paper, we adopt the *Taut-Cable Model*, which assumes the cable is rigid, massless and of length l .

In order to have a decentralized problem, we focus on deriving the individual cable tensions as functions of states of the cable, the load and the UAV without requiring direct information about the states of other UAVs. The tension equations can be derived using the relative acceleration of rigid bodies:

$$\ddot{\mathbf{p}}_i = \ddot{\mathbf{p}}_L + \mathbf{a}_{\text{centripetal}} + \mathbf{a}_{\text{tangential}} \quad (2)$$

where $\mathbf{p}_L = [x_L, y_L, z_L]^T$ denotes the load position.

Considering the tension $\mathbf{T}_i \in \mathbb{R}^3$ in the cable as a disturbance force on the UAV, we get:

$$m_i \ddot{\mathbf{p}}_i = \mathbf{F}_i - m_i g \mathbf{z}_W - \mathbf{T}_i \quad (3)$$

Substituting (3) into (2) and solving for $\mathbf{T}_i$:

$$\mathbf{T}_i = \mathbf{F}_i - m_i (\ddot{\mathbf{p}}_L + g \mathbf{z}_W) - m_i l \|\boldsymbol{\omega}_i\|^2 \boldsymbol{\xi}_i - m_i l \dot{\boldsymbol{\omega}}_i \times \boldsymbol{\xi}_i \quad (4)$$

where $\boldsymbol{\xi}_i = (\mathbf{p}_L - \mathbf{p}_i) / \|\mathbf{p}_L - \mathbf{p}_i\|$ is a unit vector in the direction of the cable, $\boldsymbol{\omega}_i \in \mathbb{R}^3$ is the angular velocity of the cable, $\dot{\boldsymbol{\omega}}_i \in \mathbb{R}^3$ is the angular acceleration of the cable, and $\times$ denotes the cross-product.

Since $\mathbf{T}_i$ and $\boldsymbol{\xi}_i$ are co-linear, we can project both sides of (4) into the direction of $\boldsymbol{\xi}_i$ to get:

$$\mathbf{T}_i = \boldsymbol{\xi}_i^T \mathbf{F}_i \boldsymbol{\xi}_i - m_i \boldsymbol{\xi}_i^T (\ddot{\mathbf{p}}_L + g \mathbf{z}_W) \boldsymbol{\xi}_i - m_i l \|\boldsymbol{\omega}_i\|^2 \boldsymbol{\xi}_i. \quad (5)$$

It is important to mention that the acceleration of the load $\ddot{\mathbf{p}}_L$ is a representation of the effect of the other cables tensions since the dynamics of the point-mass load are expressed by:

$$m_L \ddot{\mathbf{p}}_L = \sum_{i=1}^N \mathbf{T}_i - m_L g \mathbf{z}_W. \quad (6)$$

2.3 Control objectives

We assume that all the robots are able to know or estimate their altitude z_i expressed in the global frame. This assumption can be valid even without relying on a Global Positioning System (GPS) since the altitude can be estimated using other sensors like barometer. Thanks to this assumption, an altitude controller can be designed to

ensure that the group of robots are stabilized at a desired hovering altitude. With a view to ensure that the load weight is uniformly distributed for the group of robots, the desired formation will be an evenly spaced circular formation whose center is the position of the load in the $X - Y$ plane and at the desired altitude.

In order to make the load track a desired planar trajectory in 2D, our strategy is based on a leader-follower approach. We assume that there is a leader in the team that is able to control its position with a position tracking controller. The other robots in the team are followers. The followers have no information about their own positions neither the positions of their neighbors. However, we assume that each follower robot is able to measure or estimate the tension $\mathbf{T}_i$ of its' cable. The control objective is then to stabilize the robots to a desired configuration and maintain that configuration during the transportation task. We aim to achieve this control objective relying only on local information about cable tension while avoiding any kind of communication.

3. CONTROLLER DESIGN

To achieve the aforementioned control objectives, we define a desired formation of N UAVs carrying a load. We then analyze the tension relationship with the formation geometry and propose a control algorithm to maintain the desired formation during the load transportation task.

In this section, we separate the control design of the 3D thrust $\mathbf{F}_i = [F_{xi}, F_{yi}, F_{zi}]^T$ into two problems. The $X - Y$ components of the thrust $\mathbf{F}_{xyi} = [F_{xi}, F_{yi}]^T$ are designed by a 2D tension-based formation controller and F_{zi} is computed independently by an altitude controller.

3.1 Altitude controller

With the knowledge of the UAVs' mass and an estimate of the tension, we use a PID controller with a feed-forward term to stabilize each UAV at the desired altitude and compensate for the weight of the UAV and the cable tension component in z_W denoted by T_{zi} :

$$F_{zi} = m_i (K_{Pz}e_{zi} + K_{Iz} \int e_{zi} dt + K_{Dz}\dot{e}_{zi} + g) + T_{zi} \quad (7)$$

where $e_{zi} = z^{\text{ref}} - z_i$ denotes the altitude error with respect to the desired altitude reference z^{ref} , and K_{Pz} , K_{Iz} , and K_{Dz} are the control gains.

The controller gains are chosen applying a Linear Quadratic Regulator technique for the linearized model of the UAV in free flying, see Skantzikas et al. (2024) for more details.

3.2 Formation design

We consider that the desired formation is an evenly spaced circular distribution with center $\mathbf{c} = [x_L, y_L]^T$ and radius $r > 0$ placed at an altitude of z^{ref} .

Let $\mathbf{d}_i \in \mathbb{R}^2$ define the relative 2D position between UAV i and the formation center $\mathbf{c}$. The desired relative position $\mathbf{d}_i^{\text{des}}$ to obtain an evenly spaced configuration for each robot $i = 1, \dots, N$ is designed as $\mathbf{d}_i^{\text{des}} = r[\cos \alpha_i, \sin \alpha_i]^T$ where $\alpha_i = \frac{2\pi}{N}(i - 1) + \gamma$, for $\gamma \in \mathbb{R}$ any constant angle.

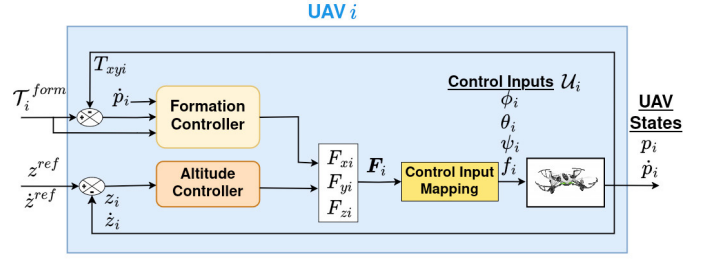


Fig. 3. Block diagram of the proposed control strategy.

The relation between $\mathbf{d}_i$ and the $X - Y$ components of the tension, denoted by $\mathbf{T}_{xyi}$, can be found using trigonometry such that:

$$\mathbf{T}_{xyi} = T_{zi} \frac{\mathbf{d}_i}{\sqrt{l^2 - \|\mathbf{d}_i\|^2}} \quad (8)$$

Considering that the UAVs are hovering while keeping the desired formation, the desired 2D tension $\mathcal{T}_i^{\text{form}} \in \mathbb{R}^2$ for each robot can be expressed as:

$$\mathcal{T}_i^{\text{form}} = T_{zi}^{\text{eq}} \frac{\mathbf{d}_i^{\text{des}}}{\sqrt{l^2 - \|\mathbf{d}_i^{\text{des}}\|^2}} \quad (9)$$

where T_{zi}^{eq} denotes the z -component of the cable tension at the equilibrium state, which occurs when the UAVs are positioned in the desired formation. This component can be calculated using $T_{zi}^{\text{eq}} = \frac{m_L g}{N}$.

3.3 Formation controller

The relation between the cables tension and the desired configuration of the UAVs at the hovering equilibrium state allows controlling the relative distance vector $\mathbf{d}_i$ in the multi-robot formation by controlling the tension of the cable. In other words, the desired evenly spaced formation can be stabilized by ensuring that each cable tension converges to its desired reference defined in (9).

Ideally, the followers have to maintain their formation tension $\mathbf{T}_{xyi} = \mathcal{T}_i^{\text{form}}$. However, due to a physical delay caused by the inertia of the load, this phenomena causes the fleet to amplify any oscillations of position of the load. In order to converge to the desired formation tension while addressing the oscillation problem, the following control law is proposed:

$$\mathbf{F}_{xyi} = K_{PT} \mathbf{e}_{Ti} + K_{DT} \dot{\mathbf{e}}_{Ti} - m_i K_v \dot{\mathbf{p}}_{xyi} + \mathcal{T}_i^{\text{form}} \quad (10)$$

where $\mathbf{e}_{Ti} = \mathcal{T}_i^{\text{form}} - \mathbf{T}_{xyi}$ is the tension error in the $X - Y$ components, K_{PT} and K_{DT} are the formation controller gains, and K_v is the velocity damping gain.

The distributed control law, shown in Fig.3, enables convergence to the desired formation tension and assists in managing any external forces acting on the load. This approach facilitates cooperative transportation of the load, driven by a *leader* that steers the formation towards a global goal. This leader can be one of the UAVs or even an external agent (for example, a human operator acting directly on the load).

3.4 Leader position controller

To achieve autonomous trajectory tracking of the load, at least one leader in the team is required. The leader

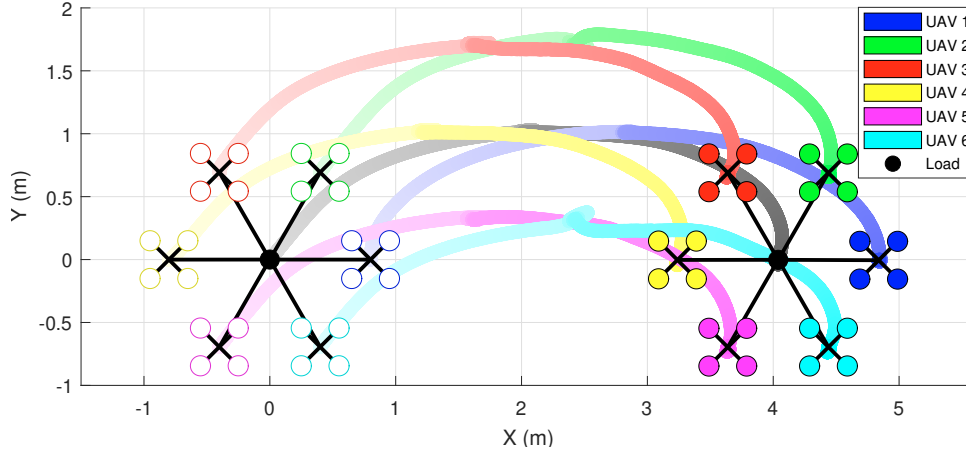


Fig. 4. Simulation results: evolution of the UAVs and load trajectories where UAV 1 (dark blue) is the leader.

must be aware of its own position, its desired position in the formation, as well as the load's desired trajectory. The leader k has a position controller to track a desired trajectory and consequently enforcing the formation to move while carrying the load. The 2D position of leader k , denoted by $\mathbf{p}_{xyk} \in \mathbb{R}^2$, is controlled by the following PID controller with a feed-forward term:

$$\mathbf{F}_{xyk} = m_k (K_{P_l} \mathbf{e}_k + K_{I_l} \int \mathbf{e}_k dt + K_{D_l} \dot{\mathbf{e}}_k) + \mathbf{T}_{xyk} \quad (11)$$

where $\mathbf{p}_{xyk}^{\text{ref}} \in \mathbb{R}^2$ is the 2D reference position for leader k , $\mathbf{e}_k = \mathbf{p}_{xyk}^{\text{ref}} - \mathbf{p}_{xyk}$ is the position error, and K_{P_l} , K_{I_l} , and K_{D_l} are the position controller gains. The reference position for each leader k is computed taking into account the desired 2D position for the load $\mathbf{p}_{xyL}^{\text{ref}}$ and its desired relative distance $\mathbf{d}_k^{\text{des}}$, such that $\mathbf{p}_{xyk}^{\text{ref}} = \mathbf{p}_{xyL}^{\text{ref}} + \mathbf{d}_k^{\text{des}}$.

3.5 Control Input mapping

Without loss of generality we assume the yaw angle $\psi_i = 0$ for all the UAVs. The control inputs $\mathbf{U}_i = [\phi_i, \theta_i, \psi_i, f_i]^T$ can be retrieved by decomposing $\mathbf{F}_i$ into a thrust magnitude and rotation angles. Here we consider the Z-Y-X Euler convention, which means that UAV i executes the rotation angles ψ_i, θ_i, ϕ_i , consequently:

$$\phi_i = \text{atan2}(F_{yi}, \sqrt{F_{zi}^2 + F_{xi}^2}) \quad (12)$$

$$\theta_i = \text{atan2}(F_{xi}, F_{zi}) \quad (13)$$

$$f_i = \|\mathbf{F}_i\| \quad (14)$$

where atan2 is the two-argument arc-tangent function which outputs an angle in the range $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

4. SIMULATION RESULTS

The simulation environment was implemented in Matlab/Simulink and includes the non-linear model of the MAMBO Parrot mini-UAVs, including a model of the closed-loop angle dynamics (see Skantzikas et al. (2024)).

4.1 Simulation Scenario

The simulation scenario presents a team of $N = 6$ UAVs evenly distributed in a circular formation of radius $r = 0.8m$ at a desired altitude $z^{\text{ref}} = 1.1m$. There is one leader (UAV 1) and five followers. The system parameters are:

$m_i = 0.0785kg \forall i, l = 1m, m_L = 0.05kg$. The control gains for the altitude controller are set to $K_{P_z} = 5.2$, $K_{I_z} = 0.13$, $K_{D_z} = 2.86$, and for the leader position controller to $K_{P_l} = 3.8$, $K_{I_l} = 0.2$, $K_{D_l} = 2.5$.

The desired trajectory of the load over time $t \in [0, 40s]$ is defined by a half-ellipse, starting at $\mathbf{p}_L^{\text{start}} = [0, 0]^T$ at $t = 10s$, a middle point $\mathbf{p}_L^{\text{middle}} = [2, 1]^T$ where the load stops from $t = 20s$ to $t = 30s$, then continues to the endpoint $\mathbf{p}_L^{\text{end}} = [4, 0]^T$ where it remains stationary.

To analyze the influence of each control gain, we studied three cases of the tension-based controller:

Case 1: $K_{P_T} = 1$, $K_{D_T} = 0.05$, $K_v = 0.2$

Case 2: $K_{P_T} = 1$, $K_{D_T} = 0.1$, $K_v = 2$

Case 3: $K_{P_T} = 2$, $K_{D_T} = 0.4$, $K_v = 2$

To better understand the simulation scenario we present Fig.4 which shows the time evolution of the 2D trajectories of all UAVs and the load. The tension-based formation control gains are those of Case 3. The 2D position of the leader UAV 1 (dark blue), is controlled by the position controller in (11). The team of UAVs is able to keep the desired evenly spaced circular formation while the leader is moving. Each UAV maintains its desired relative distance $\mathbf{d}_i$ by relaying only on the tension in the cable. The tension-based controller enforces each UAV to move following an ellipse trajectory without having any information about its 2D position neither the desired trajectories for the leader or the load.

4.2 Analysis of the results

To evaluate the overall performance of the fleet we analyze the load position and the cable tension tracking which have a direct relation with the desired formation.

The time evolution of the Y component of the cable tension T_{yi} for the follower UAV 6 under the three proposed tension-based controller cases is shown in Fig.5. In Case 1 (green), the UAV is very sensitive to the change in the tension when the leader starts moving at $t = 10$ causing it aggressively tries to converge to the desired tension. However, the tension starts oscillating when the acceleration of the load changes around $t = 20$ due to the aforementioned physical delay problem.

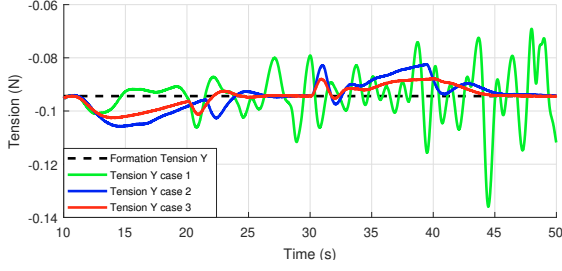


Fig. 5. Evolution of the Y tension component for UAV 6 for three different control gains (solid lines) compared to the desired formation tension (dashed black line).

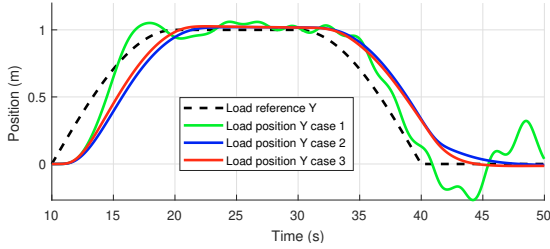


Fig. 6. Evolution of the Y position of the load for three different control gains (solid lines) compared to the desired trajectory (dashed black line).

In Case 2 (blue), a considerable reduction of the tension oscillatory behavior is achieved by increasing the damping term K_v . However this case encounters high tension error, which indicates that the UAV is resisting the motion of the load. To reduce this resistance the tension control gains in Case 3 (red) are better tuned which results in reducing the error peaks while still eliminating the oscillations.

To evaluate the overall performance of the whole fleet in the three cases, the Y component of the load position is shown in Fig.6. For Case 1 (green), the load tracks the trajectory rapidly, which is caused by the followers' rapid response to tension errors. We notice that the oscillation peaks are gradually increasing, leading the system to eventually become unstable. We notice there are almost no oscillation in both Case 2 (blue) and Case 3 (red), we also notice that Case 3 have faster reference tracking than Case 2 due to maintaining less tension error.

5. REAL-WORLD IMPLEMENTATION

The experimental setup, shown in Fig.1, consists of three MAMBO Parrot mini-UAVs attached to a load via cables (video available at: <https://youtu.be/ORG0gBFamWA>). The technical details of the platform, the motion capture system, and the control implementation are detailed in Skantzikas et al. (2024). The motion capture system provides the altitude of the UAVs and the position of the leader. The followers' 2D positions are not required.

The experimental scenario involves a team of $N = 3$ UAVs evenly distributed in a circular formation with radius $r = 0.516m$ at the desired altitude $z^{\text{ref}} = 1.1m$. There is one leader (UAV 2) and two followers. The system parameters are: $m_i = 0.0785kg \forall i$, $l = 0.9m$, $m_L = 0.0165kg$. The weight of the load is intentionally chosen to be small due to the mini-UAVs' limited thrust. The control gains for

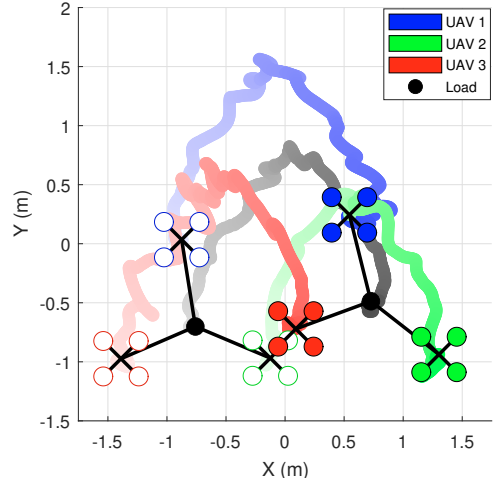


Fig. 7. Experimental results: evolution of the UAVs and load trajectories where UAV 2 (green) is the leader.

the altitude controller and the leader position controller are the same as the simulation controllers gains, and the formation controller is the same as in Case 3.

The desired trajectory of the load over time $t \in [0, 50]$ is defined as a half-ellipse, starting at $p_L^{\text{start}} = [-0.65, -0.65]^T$ at $t = 12s$, reaching $p_L^{\text{middle}} = [0, 0.65]^T$ where the load remains stationary for 10s, and finally continues to the endpoint $p_L^{\text{end}} = [0.65, -0.65]^T$.

5.1 Tension estimation

To implement the formation controller effectively, we require knowledge of the tension. However, in the absence of the appropriate force sensors, the tension of the cable cannot be directly measured. To estimate the value of the tension we exploit the motion capture system to create a virtual tension sensor.

Using the position of each UAV and the load, we can compute the direction of the cable $\xi_i = [\xi_x, \xi_y, \xi_z]^T$. Since the tension T_i is parallel to ξ_i , we can use the following geometric relation to estimate the tension:

$$T_{xyi} = \frac{T_{zi}}{\xi_z} \xi_{xy} \quad (15)$$

where $\xi_{xy} = [\xi_x, \xi_y]^T$. Note that the motion capture system of the experimental platform provides the position of an object with a certain level of accuracy. Even if in our approach, the follower robots do not rely on their positions to keep the formation, the uncertainties on position measurements will lead to errors in the estimation of the cable tension.

Although the leader must occasionally deviate from the desired formation to guide it towards the intended position. We design the trajectory to have small acceleration so that this deviation is small. Under these conditions, it is reasonable to assume that all UAVs contribute equally to lifting the load. Consequently, the Z-component of the cable tension for each UAV is calculated as $T_{zi} = \frac{m_L g}{N}$.

5.2 Analysis of the results

The time evolution of the 2D trajectories of the UAVs and the load for the experimental scenario previously described

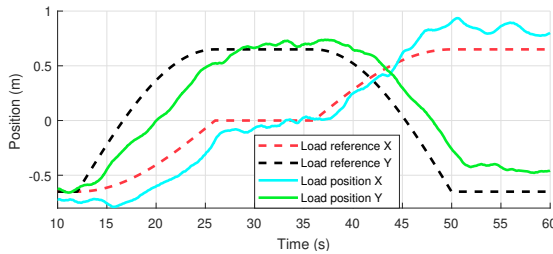


Fig. 8. Evolution of the load 2D position (solid lines) compared to the desired reference (dashed lines).

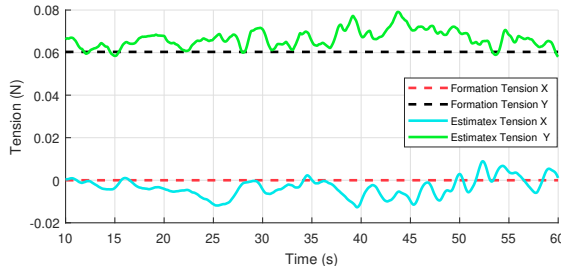


Fig. 9. Estimated tension compared with the desired tension for UAV 3.

is presented in Fig.7. The 2D position of the leader UAV 2 (green) is controlled by the position controller in (11). The tension-based controller (10) ensures that the two follower UAVs (1 and 3) keep the desired uniformly distributed formation while following a trajectory that closely matches the expected half-ellipse trajectory.

The MAMBO Parrot mini-UAVs have a limited load capacity and for the scenario presented the rotors are operating near their maximum capability. These aforementioned factors and the errors in the tension estimation lead to a more oscillatory behavior of the follower UAVs compared to the simulation results. Despite this particularly challenging situation, the team of UAVs is able to keep the formation without inter-robot communication while carrying a load. Moreover, the load follows a smooth trajectory, demonstrating the effectiveness and the potential of the proposed tension-based controller, as shown in Fig.8.

To elaborate further on the control law and estimation, Fig.9 displays the tension tracking of one of the followers (UAV 3). Both tension components oscillate around the desired formation tension. Overall, the real-world results closely match those from the simulation with a small steady state error, with major improvements anticipated using better tension and Euler angle estimations.

6. CONCLUSIONS

In this paper, we propose a distributed control strategy to stabilize a group of UAVs transporting a point-mass load. The proposed control law relies only on local information, thus eliminating the need of inter-robot communication and global position knowledge. The control algorithm integrates an altitude controller to maintain multiple aerial robots at the desired altitude and a tension-based formation controller to stabilize them to a desired 2D circular formation using cable tension feedback.

The proposed strategy is implemented within a leader-follower framework, allowing for cooperative aerial transportation of the load, guided by a leader who steers the formation along a desired trajectory. Numerical simulations and real-world experiments of the proposed control strategy have been conducted with a group of MAMBO Parrot mini-UAVs transporting a load along an elliptical trajectory. The results demonstrate successful load position tracking while the UAVs maintain the desired formation around the load.

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